

SESSION

1



# Numbers

## Learning Objectives

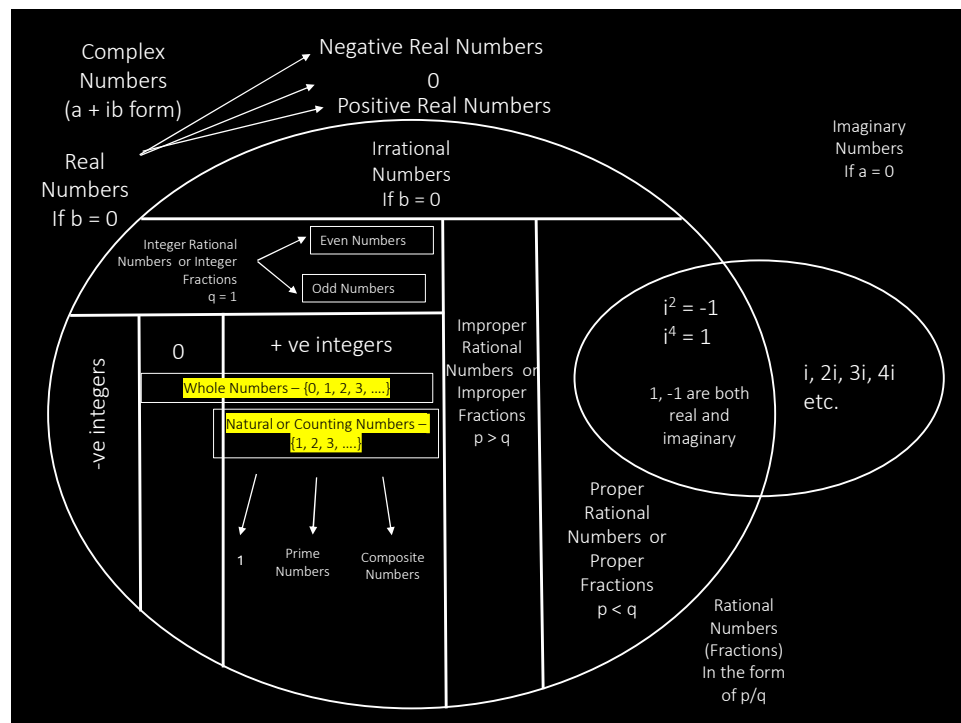
1. **Classification of Numbers**
2. **Power Cycles**
3. **Factors, Multiples**
3. **Highest Common Factors, Lowest Common Multiples**

# SESSION 1 – Numbers I

by PANKAJ BURHANPURKAR



SESSION 1 – Numbers I



## How to convert RECURRING REPEATING DECIMALS in FRACTIONS

1.  $0.33333\ldots$ 2.  $0.565656\ldots$ 3.  $0.527527527\ldots$ 4.  $0.522222\ldots$ 5.  $0.54262626\ldots$ 6.  $0.5299999\ldots$ 

SESSION 1 – Numbers I

1 of 4

## How to find PRIME NUMBERS

1. 323

2. 547

3. 239

4. 421

5. 481

SESSION 1 – Numbers I

2 of 4

## Power Cycles

1 - 1  
 2 - 2 4 8 6  
 3 - 3 9 7 1  
 4 - 4 6  
 5 - 5  
 6 - 6  
 7 - 7 9 3 1  
 8 - 8 4 2 6  
 9 - 9 1

SESSION 1 – Statistics

3 of 4



$$\begin{array}{r}
 \text{(Quotient)} \\
 5 \\
 \text{(Divisor) } 5 \overline{) 27} \text{ (Dividend)} \\
 \underline{25} \\
 2 \text{ (Reminder)}
 \end{array}$$

SESSION 1 – Numbers I

Even + Even = Even

Even + Odd = Odd

Odd + Odd = Even

$e / e$  = both integer and non-integer  
 $e / o$  = both integer and non-integer  
 $o / o$  = both integer and non-integer

**$o / e$  = always will be non-integer**

-----  
Even - Even = Even

Even - Odd = Odd

Odd - Odd = Even

-----  
Even  $\times$  Even = Even

Even  $\times$  Odd = Even

Odd  $\times$  Odd = Odd

SESSION 1 – Numbers I



# Thanks!

SESSION 1 – Numbers I

## SESSION 1 – Numbers II

by PANKAJ BURHANPURKAR



SESSION 1 – Numbers II



### Fundamental Theorem of Arithmetic

**Every composite number can be represented as the product of its prime numbers**

$$A^p \times B^q \times C^r \times D^s \dots\dots$$

Where A, B, C, D .... are prime numbers and  
p, q, r, s ... are natural numbers.

SESSION 1 – Numbers II

1 of 4

**Factor** – Factors are the numbers which completely divides a number.

**Multiple** – Multiples are the numbers which gets completely divided by a number.

1    2    3    4    6    12

12

12 24 36 48 60 72 84 96 108 120  
132 144 156 168 180 192 204.....

1    2    4    5    10    20

20

20 40 60 80 100 120 140 160 180  
200 220 240 260 280 300 320 340.....

Factors

Multiples

SESSION 1 – Numbers II

1 of 4

### Important Results

Number of Factors (F) =  $(p+1)(q+1)(r+1)....$

Product of all the positive Factors =  $N^{F/2}$

Sum of all the positive Factors =  $\frac{A^{p+1}-1}{A-1} \cdot \frac{A^{q+1}-1}{B-1} \cdot \frac{C^{r+1}-1}{C-1} ....$

Multiples forms AP, so use AP results to find Number of multiples, Sum and Product

Sum of multiples =  $N \cdot n(n+1)/2$

Product of multiples =  $N^n \cdot n!$

To find n – compare last multiple to  $N + (n-1)N$

SESSION 1 – Numbers II

1 of 4

**Factor** – Factors are the numbers which completely divides a number.

**Multiple** – Multiples are the numbers which gets completely divided by a number.

1    2    3    4    6    12

12

12 24 36 48 60 72 84 96 108 120  
132 144 156 168 180 192 204.....

1    2    4    5    10    20

20

20 40 60 80 100 120 140 160 180  
200 220 240 260 280 300 320 340.....

Highest Common Factor

Lowest Common Multiple

SESSION 1 – Numbers II

1 of 4

$$\begin{array}{r}
 \text{(Quotient)} \\
 5 \\
 \hline
 \text{(Divisor)} 5 \overline{) 27} \text{ (Dividend)} \\
 \underline{25} \\
 2 \text{ (Reminder)}
 \end{array}$$

HCF = GCD (Greatest Common Divisor)

LCM = LCD (Lowest Common Dividend)

SESSION 1 – Numbers II

1 of 4

Find the smallest number which when divided by 4 or 7 leaves a remainder of 3 in each case and the number being greater than the two divisors.

SESSION 1 – Numbers II

1 of 14

Find the largest three-digit number which when divided by 3 or 8 leaves a remainder of 2 in each case?

SESSION 1 – Numbers II

2 of 14

Find the smallest six-digit number, which when divided by 3 or 11 leaves a remainder of 2 in each case.

SESSION 1 – Numbers II

3 of 14

Find the smallest number greater than the divisors which when divided by 4, 7 or 11 leaves a remainder of 3 in each case.

SESSION 1 – Numbers II

4 of 14

Find the smallest number which when divided by 4 and 7 gives remainders of 2 and 5 respectively.

SESSION 1 – Numbers II

5 of 14

Find the largest four digit number which when divided by 5 and 9 gives a remainder of 3 and 7 respectively.

SESSION 1 – Numbers II

6 of 14

Find the smallest six digit number which when divided by 7 leaves a remainder of 3 and when divided by 11 leaves a remainder of 7.

SESSION 1 – Numbers II

7 of 14

Find the smallest number which when divided by 5, 7, 11 and 13 leaves respective remainders of 2, 4, 8 and 10.

SESSION 1 – Numbers II

8 of 14

Find out the smallest number which when divided by 7 gives a remainder of 3 and when divided by 5 gives the remainder of 2.

SESSION 1 – Numbers II

9 of 14

Find the smallest number which when divided by 7 leaves a remainder of 6 and when divided by 11 leaves a remainder of 8.

SESSION 1 – Numbers II

10 of 14

Find the largest number with which when 72 and 119 are divided respective remainders of 3 and 4 are left.

SESSION 1 – Numbers II

11 of 14

Find the largest number with which when 906 and 650 are divided they leave respective remainders of 3 and 5.

SESSION 1 – Numbers II

12 of 14

Find the largest number with which when 555, 1275 and 1635 are divided, the remainders are the same.

SESSION 1 – Numbers II

13 of 14

Find the largest number with which when 437, 857 and 1557 are divided, leaves the same remainder in each case.

SESSION 1 – Numbers II

14 of 14



**Thanks!**

## Class Practice Exercise 01

### Classification of Numbers, Power Cycles

**DIRECTIONS:** For the following questions, four options are given. Choose the correct answer.

1. Which of the following numbers is an irrational number?  
 (1)  $\frac{22}{7}$  (2)  $\sqrt{16}$   
 (3)  $2\sqrt{2}$  (4) 3.5
2. The sum of two odd numbers is:  
 (1) Odd (2) Even  
 (3) Prime (4) Composite
3. Which of the following is a prime number?  
 (1) 4 (2) 6  
 (3) 9 (4) 11
4. A number that is divisible by more than two numbers is called:  
 (1) Prime (2) Composite  
 (3) Even (4) Odd
5. Which of the following fractions is an improper fraction?  
 (1)  $\frac{3}{4}$  (2)  $\frac{5}{3}$   
 (3)  $\frac{2}{5}$  (4)  $\frac{1}{2}$
6. The smallest natural number is:  
 (1) 0 (2) 1  
 (3) 2 (4) -1
7. Which of the following pairs are co-prime numbers?  
 (1) (14, 21) (2) (15, 28)  
 (3) (18, 24) (4) (35, 49)
8. The decimal representation of a rational number:  
 (1) Terminates or repeats  
 (2) Is non-terminating and non-repeating  
 (3) Is always an integer  
 (4) None of the above
9. Which of the following is an autobiographical number?  
 (1) 1210 (2) 1234  
 (3) 2020 (4) 1111
10. The product of two negative integers is:  
 (1) Positive (2) Negative  
 (3) Zero (4) None of above
11. Which of the following numbers is both real and imaginary?  
 (1) 0 (2) 2  
 (3) i (4) -1
12. The sum of the first five prime numbers is:  
 (1) 18 (2) 26  
 (3) 28 (4) 33
13. Which of the following is a proper fraction?  
 (1)  $\frac{7}{3}$  (2)  $\frac{5}{5}$   
 (3)  $\frac{2}{3}$  (4)  $\frac{9}{4}$
14. If the sum of two numbers is even and their product is odd, then which of the following is true?  
 (1) Both numbers are even  
 (2) Both numbers are odd  
 (3) One number is even, and the other is odd  
 (4) None of the above
15. The smallest prime number greater than 100 is:  
 (1) 101 (2) 103  
 (3) 107 (4) 113
16. The product of an even number and an odd number is always:  
 (1) Even (2) Odd  
 (3) Prime (4) Composite
17. The reciprocal of an improper fraction is always:  
 (1) A proper fraction  
 (2) An improper fraction  
 (3) An integer  
 (4) A mixed fraction
18. What is the only even prime number?  
 (1) 1 (2) 2  
 (3) 3 (4) 4
19. The number 0 is classified as:  
 (1) A whole number  
 (2) A natural number  
 (3) A prime number  
 (4) A Irrational number
20. A number that is divisible by both 2 and 3 is always divisible by:  
 (1) 5 (2) 6  
 (3) 8 (4) 12
21. Which of the following is an irrational number?  
 (1)  $\frac{4}{5}$  (2)  $7\sqrt{7}$   
 (3) 3.1416 (4)  $\frac{22}{7}$
22. The sum of any two co-prime numbers is always:  
 (1) Even (2) Odd  
 (3) Prime (4) None of above

23. Which of the following is an example of a composite number?  
 (1) 13 (2) 17  
 (3) 25 (4) 19
24. A prime number has exactly:  
 (1) One factor (2) Two factors  
 (3) Three factors (4) Infinite factors
25. Which of the following is NOT an integer?  
 (1) -5 (2) 0  
 (3) 3.5 (4) 100
26. Which of the following numbers is an autobiographical number?  
 (1) 1210 (2) 4321  
 (3) 1111 (4) 2020
27. The sum of the first four odd natural numbers is:  
 (1) 16 (2) 10  
 (3) 20 (4) 12
28. If a number ends in 0 or 5, then it is divisible by:  
 (1) 2 (2) 3  
 (3) 5 (4) 7
29. The sum of three consecutive integers is always:  
 (1) Even (2) Odd  
 (3) Prime (4) Divisible by 3
30. The number 100 is a:  
 (1) Prime number  
 (2) Composite number  
 (3) Whole number  
 (4) Both (2) and (3)
31. Which of the following is a prime number?  
 (1) 91 (2) 81  
 (3) 79 (4) 49
32. Which of the following is an odd composite number?  
 (1) 11 (2) 17  
 (3) 21 (4) 29
33. What is the smallest two-digit prime number?  
 (1) 10 (2) 11  
 (3) 12 (4) 13
34. The sum of any two even numbers are always:  
 (1) Even (2) Odd  
 (3) Prime (4) Composite
35. Which of the following is a co-prime pair?  
 (1) 12, 15 (2) 14, 21  
 (3) 18, 25 (4) 20, 30
36. The decimal expansion of every rational number is either:  
 (1) Terminating (2) Recurring  
 (3) Both (1) and (2) (4) None of these
37. The greatest common divisor (GCD) of two co-prime numbers is:  
 (1) 0 (2) 1  
 (3) 2 (4) Their product
38. Which of the following is NOT a perfect square?  
 (1) 49 (2) 81  
 (3) 144 (4) 98
39. The sum of all even numbers from 1 to 10 is:  
 (1) 20 (2) 25  
 (3) 30 (4) 40
40. The product of two odd numbers is always:  
 (1) Odd (2) Even  
 (3) Prime (4) None of these
41. Which number is neither prime nor composite?  
 (1) 0 (2) 1  
 (3) 2 (4) Both (1) and (2)
42. The LCM of two prime numbers is:  
 (1) Their product (2) Their sum  
 (3) 1 (4) 0
43. Which of these is an improper fraction?  
 (1)  $\frac{3}{4}$  (2)  $\frac{7}{3}$   
 (3)  $\frac{5}{6}$  (4)  $\frac{9}{10}$
44. The largest two-digit prime number is:  
 (1) 89 (2) 97  
 (3) 99 (4) 91
45. The smallest prime number greater than 50 is:  
 (1) 51 (2) 53  
 (3) 55 (4) 57
46. If a number is divisible by both 4 and 6, then it must be divisible by:  
 (1) 12 (2) 18  
 (3) 24 (4) 36
47. The smallest positive integer is:  
 (1) 0 (2) 1  
 (3) -1 (4) 2
48. The multiplicative reciprocal of 1 is:  
 (1) 1 (2) 0  
 (3) -1 (4) Undefined
49. What is the unit digit of  $7^{100}$ ?  
 (1) 1 (2) 7  
 (3) 9 (4) 3
50. The sum of all prime numbers between 1 and 10 is:  
 (1) 15 (2) 17  
 (3) 20 (4) 25

51. What is the unit digit of  $2^{15}$   
 (1) 2 (2) 4  
 (3) 6 (4) 8
52. Find the unit digit of  $7^{20}$   
 (1) 1 (2) 3  
 (3) 7 (4) 9
53. What is the remainder when  $3^{25}$  is divided by 10?  
 (1) 3 (2) 7  
 (3) 9 (4) 1
54. Determine the unit digit of  $4^{13}$   
 (1) 2 (2) 4  
 (3) 6 (4) 8
55. What is the unit digit of  $5^{17}$ ?  
 (1) 0 (2) 2  
 (3) 5 (4) 7
56. Find the unit digit of  $6^{22}$   
 (1) 2 (2) 6  
 (3) 8 (4) 0
57. What is the remainder when  $8^{35}$  is divided by 10?  
 (1) 2 (2) 4  
 (3) 6 (4) 8
58. Find the unit digit of  $9^{21}$   
 (1) 1 (2) 3  
 (3) 7 (4) 9
59. Determine the unit digit of  $12^{25}$ .  
 (1) 2 (2) 4  
 (3) 6 (4) 8
60. What is the unit digit of  $19^{17}$ ?  
 (1) 1 (2) 3  
 (3) 7 (4) 9
61. Find the unit digit of  $27^{14}$   
 (1) 1 (2) 3  
 (3) 7 (4) 9
62. What is the remainder when  $34^{23}$  is divided by 10?  
 (1) 4 (2) 6  
 (3) 8 (4) 2
63. Find the unit digit of  $73^{21}$   
 (1) 1 (2) 3  
 (3) 7 (4) 9
64. Determine the unit digit of  $81^{32}$   
 (1) 1 (2) 3  
 (3) 7 (4) 9
65. Find the unit digit of  $99^{10}$   
 (1) 1 (2) 3  
 (3) 7 (4) 9
66. Find the remainder when  $123^{45}$  is divided by 10.

- (1) 1 (2) 3  
 (3) 7 (4) 9
67. What is the unit digit of  $37^{100}$ ?  
 (1) 1 (2) 3  
 (3) 7 (4) 9
68. Determine the unit digit of  $44^{99}$ .  
 (1) 2 (2) 4  
 (3) 6 (4) 8
69. Find the unit digit of  $52^{112}$   
 (1) 2 (2) 4  
 (3) 6 (4) 8
70. What is the unit digit of  $67^{49}$   
 (1) 1 (2) 3  
 (3) 7 (4) 9
71. What is the remainder when  $88^{77}$  is divided by 10?  
 (1) 2 (2) 4  
 (3) 6 (4) 8
72. Find the remainder when  $137^{100} - 92^{31}$  is divided by 10.  
 (1) 9 (2) 3  
 (3) 7 (4) 9
73. Determine the unit digit of  $999^{333} \times 121^{88}$   
 (1) 1 (2) 3  
 (3) 7 (4) 9
74. Find the remainder when  $(246^{145} + 528^{79}) 298^{256}$  is divided by 10.  
 (1) 2 (2) 4  
 (3) 6 (4) 8
75. Find the unit digit of  $763^{223} + 951^{199}$ .  
 (1) 2 (2) 8  
 (3) 7 (4) 9
76. Which of the following is not a Prime Number?  
 (1) 313 (2) 111  
 (3) 401 (4) 479
77. Which of the following is a Prime Number?  
 (1) 117 (2) 259  
 (3) 357 (4) 389
78. Which of the following is not a Prime Number?  
 (1) 141 (2) 331  
 (3) 211 (4) 199
79. Which of the following is a Prime Number?  
 (1) 177 (2) 231  
 (3) 351 (4) 199

80. What is the fractional value of  $0.47474747\dots$ ?

- |              |             |
|--------------|-------------|
| (1) $47/100$ | (2) $47/99$ |
| (3) $47/90$  | (4) $47/09$ |

81. What is the fractional value of  $0.356356356356\dots$ ?

- |                |               |
|----------------|---------------|
| (1) $356/1000$ | (2) $356/999$ |
| (3) $356/900$  | (4) $356/90$  |

82. What is the fractional value of  $0.4555555\dots$ ?

- |              |             |
|--------------|-------------|
| (1) $45/100$ | (2) $45/90$ |
| (3) $41/90$  | (4) $45/99$ |

83. What is the fractional value of  $0.3577777\dots$ ?

- |               |               |
|---------------|---------------|
| (1) $35/900$  | (2) $357/900$ |
| (3) $357/999$ | (4) $322/900$ |

84. What is the fractional value of  $0.743743743\dots$ ?

- |                |               |
|----------------|---------------|
| (1) $743/900$  | (2) $743/999$ |
| (3) $743/1000$ | (4) None      |

## Class Practice Exercise - 02

### Factors, Multiples, HCF and LCM I

**DIRECTIONS:** For the following questions, four options are given. Choose the correct answer.

1. If  $N = 2^4 \times 3^3 \times 5^2$ , how many positive factors does N have?
 

|        |        |
|--------|--------|
| (1) 60 | (2) 45 |
| (3) 50 | (4) 48 |
2. What is the sum of all positive factors of 360?
 

|          |          |
|----------|----------|
| (1) 1170 | (2) 1080 |
| (3) 1440 | (4) 1120 |
3. The product of all positive divisors of 180 is:
 

|             |                |
|-------------|----------------|
| (1) $180^4$ | (2) $180^6$    |
| (3) $180^9$ | (4) $180^{10}$ |
4. If  $N = p^a \times q^b \times r^c$ , then the sum of all factors of N is given by:
 

|                               |                                                                 |
|-------------------------------|-----------------------------------------------------------------|
| (1) $(a+1)(b+1)(c+1)$         | (2) $\frac{(p^{a+1}-1)(q^{b+1}-1)(r^{c+1}-1)}{(p-1)(q-1)(r-1)}$ |
| (3) $p^{a+1} q^{b+1} r^{c+1}$ | (4) None of these                                               |
5. The number of even factors of 1200 is:
 

|        |        |
|--------|--------|
| (1) 25 | (2) 30 |
| (3) 28 | (4) 24 |
6. The HCF and LCM of two numbers are 15 and 840, respectively. If one number is 105, what is the other number?
 

|         |         |
|---------|---------|
| (1) 120 | (2) 140 |
| (3) 125 | (4) 135 |
7. The product of two numbers is 2800, and their HCF is 7. What is their LCM?
 

|         |         |
|---------|---------|
| (1) 400 | (2) 450 |
| (3) 300 | (4) 350 |
8. The greatest number that will divide 58, 114, and 152 leaving remainders 4, 6, and 8 respectively is:
 

|        |        |
|--------|--------|
| (1) 12 | (2) 16 |
| (3) 18 | (4) 20 |
9. If the LCM of two numbers is 495 and their HCF is 5, then the possible pair of numbers is:
 

|             |            |
|-------------|------------|
| (1) 45, 55  | (2) 25, 99 |
| (3) 15, 165 | (4) 33, 75 |
10. The HCF of  $\frac{5}{6}$ ,  $\frac{10}{9}$ ,  $\frac{15}{12}$  is:
 

|                    |                    |
|--------------------|--------------------|
| (1) $\frac{5}{36}$ | (2) $\frac{5}{18}$ |
| (3) $\frac{5}{24}$ | (4) $\frac{5}{12}$ |
11. The LCM of  $\frac{3}{4}$ ,  $\frac{5}{6}$ ,  $\frac{7}{8}$  is:
 

|                      |                     |
|----------------------|---------------------|
| (1) $\frac{105}{2}$  | (2) $\frac{105}{8}$ |
| (3) $\frac{105}{24}$ | (4) $\frac{105}{4}$ |
12. The HCF of  $\frac{8}{21}$ ,  $\frac{12}{35}$ ,  $\frac{16}{49}$  is:
 

|                     |                     |
|---------------------|---------------------|
| (1) $\frac{4}{147}$ | (2) $\frac{2}{245}$ |
| (3) $\frac{8}{245}$ | (4) $\frac{4}{735}$ |
13. The LCM of three fractions is obtained by:
 

|                                                              |                                                      |
|--------------------------------------------------------------|------------------------------------------------------|
| (1) LCM of numerators divided by HCF of denominators         | (2) HCF of numerators divided by LCM of denominators |
| (3) Product of numerators divided by product of denominators | (4) None of these                                    |
14. The traffic lights at three different road crossings change every 40 sec, 60 sec, and 72 sec, respectively. If they change simultaneously at 8:20 AM, at what time will they change together again?
 

|                |                |
|----------------|----------------|
| (1) 8:27:12 AM | (2) 8:30:00 AM |
| (3) 8:24:00 AM | (4) 8:26:00 AM |
15. Three bells toll together at intervals of 9, 12, and 15 seconds. How many times will they toll together in an hour?
 

|        |        |
|--------|--------|
| (1) 20 | (2) 21 |
| (3) 19 | (4) 22 |
16. Find the largest number that divides 125, 250, and 375 leaving remainders 5, 10, and 15 respectively.
 

|         |         |
|---------|---------|
| (1) 120 | (2) 125 |
| (3) 130 | (4) 135 |
17. The sum of two numbers is 495. If their HCF is 15 and their LCM is 2805, find the numbers.
 

|              |              |
|--------------|--------------|
| (1) 105, 390 | (2) 165, 330 |
| (3) 195, 300 | (4) 150, 345 |
18. The smallest number which when divided by 8, 12, and 20 leaves a remainder of 5 in each case is:
 

|         |         |
|---------|---------|
| (1) 235 | (2) 245 |
| (3) 250 | (4) 255 |
19. If  $N = 12!$  Then how many trailing zeros does N have?
 

|       |       |
|-------|-------|
| (1) 2 | (2) 3 |
| (3) 4 | (4) 5 |
20. The number 540 has how many perfect square factors?
 

|        |        |
|--------|--------|
| (1) 6  | (2) 8  |
| (3) 10 | (4) 12 |

21. Find the number of divisors of 720 that are perfect cubes.  
 (1) 3 (2) 4  
 (3) 5 (4) 6
22. The sum of all odd factors of 9000 is:  
 (1) 5460 (2) 4560  
 (3) 6540 (4) 5640
23. The product of all divisors of 1080 is given by:  
 (1)  $1080^{14}$  (2)  $1080^{15}$   
 (3)  $1080^{16}$  (4)  $1080^{16}$
24. If two numbers are in the ratio 3:5 and their HCF is 12, then the numbers are:  
 (1) 36, 60 (2) 18, 30  
 (3) 24, 40 (4) 30, 50
25. The least number which when divided by 24, 36, and 54 leaves remainder 21 in each case is:  
 (1) 241 (2) 249  
 (3) 261 (4) 237
26. The largest number that divides 279 and 387 leaving remainders 6 and 9 respectively is:  
 (1) 12 (2) 15  
 (3) 18 (4) 21
27. If the HCF of  $\frac{3}{4}$ ,  $\frac{5}{6}$ , and  $\frac{7}{8}$  is  $x$ , then  $x$  is:  
 (1)  $\frac{1}{24}$  (2)  $\frac{1}{48}$   
 (3)  $\frac{1}{12}$  (4)  $\frac{1}{36}$
28. The LCM of  $\frac{2}{3}$ ,  $\frac{4}{5}$ ,  $\frac{6}{7}$  is:  
 (1)  $\frac{24}{1}$  (2)  $\frac{12}{1}$   
 (3)  $\frac{48}{1}$  (4)  $\frac{36}{1}$
29. If the HCF of two numbers is 5 and their LCM is 180, find the numbers given that their sum is 65.  
 (1) 25, 40 (2) 35, 30  
 (3) 45, 20 (4) 50, 15
30. The sum of two numbers is 420. Their LCM is 900, and HCF is 30. Find the numbers.  
 (1) 150, 270 (2) 120, 300  
 (3) 180, 240 (4) 210, 210
31. The smallest number that when increased by 10 is exactly divisible by 12, 18, and 24 is:  
 (1) 58 (2) 56  
 (3) 62 (4) 70
32. The bells in a temple toll at intervals of 6, 9, and 12 minutes. If they toll together at 8:00 AM, when will they toll together again?  
 (1) 8:36 AM (2) 8:48 AM  
 (3) 8:54 AM (4) 9:00 AM
33. Find the greatest number that will divide 135, 225, and 315 leaving the same remainder in each case.  
 (1) 90 (2) 45  
 (3) 60 (4) 75
34. Find the smallest number which when divided by 7, 9, and 11 leaves remainders 2, 4, and 6 respectively.  
 (1) 691 (2) 693  
 (3) 697 (4) 688
35. The product of two numbers is 2028 and their LCM is 169. Find their HCF.  
 (1) 10 (2) 12  
 (3) 14 (4) 18
36. A school bus arrives at a stop every 40 minutes. Another bus arrives at the same stop every 45 minutes. If both arrived at 7:30 AM, when will they arrive together again?  
 (1) 12:30 PM (2) 1:30 PM  
 (3) 3:30 PM (4) 5:30 PM
37. The HCF of three numbers is 6, and their LCM is 360. If one of the numbers is 12, find the other two numbers.  
 (1) 18, 20 (2) 24, 90  
 (3) 15, 36 (4) 20, 24
38. If  $x$ ,  $y$ ,  $z$  are three numbers such that  $\text{HCF}(x, y) = 2$ ,  $\text{HCF}(y, z) = 3$ , and  $\text{HCF}(x, z) = 5$ , then the possible value of  $\text{LCM}(x, y, z)$  is:  
 (1) 10 (2) 20  
 (3) 30 (4) 50
39. Find the LCM of the first 10 natural numbers.  
 (1) 2520 (2) 5040  
 (3) 20160 (4) 10080
40. Find the largest number which exactly divides  $100!$ ,  $200!$ , and  $300!$ .  
 (1)  $100!$  (2)  $200!$   
 (3)  $300!$  (4) None of these
41. The number 108 is multiplied by the smallest number to make it a perfect cube. Find the number.  
 (1) 2 (2) 3  
 (3) 4 (4) 6
42. Find the largest number that divides 3050 and 4650 leaving the remainder 50 in both cases.  
 (1) 200 (2) 250  
 (3) 300 (4) 350

43. Find the smallest number which when divided by 16, 18, 20, and 25 leaves a remainder of 4 in each case.

|     |      |     |      |
|-----|------|-----|------|
| (1) | 3604 | (2) | 3606 |
| (3) | 3608 | (4) | 3612 |

44. Find the LCM of the first 8 odd natural numbers.

|     |       |     |       |
|-----|-------|-----|-------|
| (1) | 45045 | (2) | 40540 |
| (3) | 37835 | (4) | 51350 |

45. The product of two numbers is 1386 and their sum is 75. Find the LCM.

|     |     |     |     |
|-----|-----|-----|-----|
| (1) | 462 | (2) | 277 |
| (3) | 315 | (4) | 294 |

46. The HCF of  $7^4$ ,  $7^6$ ,  $7^8$  is:

|     |       |     |       |
|-----|-------|-----|-------|
| (1) | $7^4$ | (2) | $7^6$ |
| (3) | $7^8$ | (4) | $7^2$ |

**DIRECTIONS:** Each of the questions 1 to 8 consists of two quantities, one in column A and one in column B. You are to compare two quantities and choose

- (1) if quantity in column A is greater;
- (2) if quantity in column B is greater;
- (3) if quantities in both the columns are equal;
- (4) if the relationship cannot be determined from the information given.

47.  $N = P^x$ , P is prime and N is perfect square

**Quantity A:** x

**Quantity B:** 3

48. **Quantity A:**  $80! \times 80$

**Quantity B:**  $81! - 80!$

49.  $Y = x + x + x + \dots$  x times,  $Y < 200$ , x is a positive integer

**Quantity A:** Number of values Y can take

**Quantity B:** 14

50. Sum of 25 natural numbers is odd. If there are M even numbers out of 25

**Quantity A:** Number of values M can take

**Quantity B:** 12

51. **Quantity A:** Remainder when 4172 is divided by 16

**Quantity B:** Remainder when  $922 \times 117^3$  is divided by 15

52. Four digits of the number 29138576 are omitted so that the result is as large as possible.

**Quantity A:** The largest omitted digit

**Quantity B:** 6

**DIRECTIONS:** Each of the questions 9 to 17 are multiple choice questions. Solve them individually to ascertain the best possible option.

53. Sum of 2 natural numbers is 144 and their HCF is 24. Find the number of positive factors of the larger of the 2 numbers?

|     |    |     |    |
|-----|----|-----|----|
| (1) | 60 | (2) | 16 |
| (3) | 8  | (4) | 24 |

54. What is the minimum number of square marbles required to tile a floor of length 5 metres 78 cm and width 3 metres 74 cm?

|     |     |     |     |
|-----|-----|-----|-----|
| (1) | 176 | (2) | 187 |
| (3) | 540 | (4) | 748 |

55. Three gold coins of weight 780 gm, 840 gm and 960 gm are cut into small pieces, all of which have equal weight. Each piece must be as heavy as possible. If one such piece is shared among two persons, then how many persons are needed to give all the pieces of gold coins?

|     |    |     |    |
|-----|----|-----|----|
| (1) | 86 | (2) | 70 |
| (3) | 43 | (4) | 35 |

### Class Practice Exercise - 03

#### Factors, Multiples, HCF and LCM II

**DIRECTIONS:** For the following questions, four options are given. Choose the correct answer.

1. Find the number of positive factors of smallest three-digit number, which is multiple of 3, 4 and 7  
(1) 5 (2) 84  
(3) 3 (4) 16
2. In how many ways 1000 can be written as a sum of two or more consecutive natural numbers?  
(1) 0 (2) 1  
(3) 2 (4) 3
3. 97792a6940b is divisible by 55. Then how many values 'a' can take?  
(1) 0 (2) 1  
(3) 2 (4) 3
4. Let P be the product of the first 30 multiples of 30. Let N be the number of factors of P. In how many ordered pairs (X, Y) can N be written as  $N = XY$  with  $\gcd(X, Y) = 19$ ?  
(1) 4 (2) 6  
(3) 8 (4) 12
5. How many times one needs to press the key, if one has to type page numbers of a book with 252 pages.  
(1) 444 (2) 999  
(3) 200 (4) 648
6. To find the GCD of n positive integers using the function  $h(a, b) = \gcd(a, b)$ , what is the minimum number of times h must be applied?  
(1)  $1/2n$  (2)  $(n - 1)$   
(3) n (4) None of these
7. On a 5-inch scale, each one-inch mark is divided into fourth with a blue mark, and each inch is also divided into eighth with a red mark. At how many places will a blue mark and red mark overlap?  
(1) 19 (2) 21  
(3) 23 (4) 25
8. The LCM of three positive integers X, Y and Z is 1192. Find the total number of ordered triplets (X, Y and Z).  
(1) 361 (2) 400  
(3) 225 (4) 259
9. Natural number N is divisible by 143 and 66. Which of the following must be true? (Mark all correct answers)  
(1) N is divisible by 8  
(2) N is divisible by 14  
(3) N is divisible by 39  
(4) N is divisible by 132  
(5) N is divisible by 286  
(6) N is divisible by 242
10. Sumit wrongly took the LCM of three distinct integers A, B, C > 1 as 840. 840 is a common multiple, but not the true LCM. Given  $\gcd(A, B, C) = 1$ , what is the maximum possible value of  $A + B + C$ ?  
(1) 427 (2) 489  
(3) 563 (4) 592
11. When a natural number N is divided by 200 remainder is 24. Which of the following must be the integers?  
(1)  $N/4$  (2)  $N/5$   
(3)  $N/11$  (4)  $N/13$
12. Three VPs visit on fixed cycles: HR every 2 days gap, Operations every 3, Sales every 5. All meet together only when their visits coincide. CEO last met them on Jan 3, 2012 and is away Jan 5–28. When is their next simultaneous visit?  
(1) Feb 16, 2012 (2) Feb 19, 2012  
(3) Feb 8, 2012 (4) Feb 12, 2012
13. Consider  $N = 2^5 \times 3^4 \times 5^3$ . Find the total number of factors of "N".  
(1) 120 (2) 136  
(3) 160 (4) 124
14. LCM of two numbers is  $ab^3$ , where a and b are prime numbers. If one of the numbers is  $b^3$ , then which of the following cannot be the other number?  
(1)  $ab^3$  (2) ab  
(3)  $ab^2$  (4) None of these
15. Consider  $N = 2^5 \times 3^4 \times 5^3$ . Find the total number of factors of "N" which are odd.  
(1)  $20^*$  (2) 36  
(3) 16 (4) 24
16. Let ab and ba be two-digit numbers where a and b are distinct digits from 0 to 9. Difference of ab and ba is HCF of ab and ba. What is the value of  $a + b$ ?  
(1) 8 (2) 9  
(3) 6 (4) 7
17. Consider  $N = 2^5 \times 3^4 \times 5^3$ . Find the total number of factors of "N" which are even.  
(1) 100 (2) 136  
(3) 116 (4) 124
18. The sum of two numbers is 462 and their HCF is 22. What is the maximum possible number of pairs that satisfy these conditions?  
(1) 6 (2) 8  
(3) 2 (4) 3
19. Consider  $N = 2^5 \times 3^4 \times 5^3$ . Find the total number of factors of "N" which are composite.  
(1) 112 (2) 136  
(3) 116 (4) 124

20. The HCF of two natural numbers is 119 and their sum is 833. What are the numbers, if they differ by the minimum possible amount?  
 (1) 119 and 714 (2) 357 and 476  
 (3) 119 and 545 (4) 238 and 545
21. Consider  $N = 2^5 \times 3^4 \times 5^3$ . Find the total number of factors of "N" which are multiple of 5.  
 (1) 80 (2) 60  
 (3) 90 (4) 100
22. The LCM of two numbers is 168 and their HCF is 28. If the sum of the numbers is 140, then the numbers are  
 (1) 112 and 28 (2) 84 and 56  
 (3) 106 and 34 (4) 56 and 28
23. Consider  $N = 2^5 \times 3^4 \times 5^3$ . Find the total number of factors of "N" which are multiple of 15.  
 (1) 72 (2) 36  
 (3) 16 (4) 24
24. Three neon signs flash every 1.5 s, 2.5 s, and 3.5 s. Each stays ON for 1 s. All flash together at  $t = 0$ . When is the next time all three are ON at the same moment?  
 (1) 26.25 s (2) 52.5 s  
 (3) 105 s (4) 210 s
25. Consider  $N = 2^5 \times 3^4 \times 5^3$ . Find the total number of factors of "N" which are perfect square.  
 (1) 14 (2) 16  
 (3) 18 (4) 20
26. A is the set of positive integers such that when divided by 2, 3, 4, 5, 6 leaves the remainders 1, 2, 3, 4, 5 respectively. How many integers between 0 and 100 belong to set A?  
 (1) 0 (2) 1  
 (3) 2 (4) None of these
27. Consider  $N = 2^5 \times 3^4 \times 5^3$ . Find the total number of factors of "N" which are perfect cube.  
 (1) 6 (2) 8  
 (3) 2 (4) 3
28. Find the greatest four-digit number which when divided by 4, 5, 6 and 7 leaves remainders 2, 3, 4 and 5 respectively.  
 (1) 9658 (2) 9662  
 (3) 9242 (4) 9982
29. Consider  $N = 2^5 \times 3^4 \times 5^3$ . Find the total number of factors of "N" which has unit's digit zero.  
 (1) 60 (2) 75  
 (3) 50 (4) 120
30. The least number which on division by 35 leaves the remainder 25 and on division by 45 leaves the remainder 35 and on division by 55 leaves the remainder 45 is  
 (1) 3455 (2) 3465  
 (3) 3455 (4) 3155
31. Consider  $N = 2^5 \times 3^4 \times 5^3$ . Find the total number of factors of "N" who's last two digits zero.  
 (1) 20 (2) 40  
 (3) 60 (4) 80
32. Find the least number which when divided by 6, 15 and 17 always leaves a remainder 1, but when divided by 7 leaves no remainder.  
 (1) 1021 (2) 86  
 (3) 211 (4) 511
33. Consider  $N = 2^5 \times 3^4 \times 5^3$ . Find the sum of all factors of "N".  
 (1) 145,612 (2) 160,372  
 (3) 175,984 (4) 1189188
34. Form five-digit strings with 1–5 used once each. Exactly two of the 1st, 3rd, and 5th spots must contain odd digits. Find the total of the digits appearing in the last spot over all such strings.  
 (1) 92 (2) 328  
 (3) 192 (4) 216
35. Consider  $N = 2^5 \times 3^4 \times 5^3$ . Find the product of all factors of "N".  
 (1)  $2^{300} \cdot 3^{240} \cdot 5^{180}$  (2)  $2^{600} \cdot 3^{480} \cdot 5^{360}$   
 (3) 4505010 (4)  $N^{60}$
36. Let a, b, c, d be four different non-zero digits. They satisfy:  $b + c + d \equiv 0 \pmod{9}$ ,  $a + c + d \equiv 0 \pmod{8}$ ,  $a + b + d \equiv 0 \pmod{5}$ ,  $a + b + c \equiv 0 \pmod{23}$ . Find ab.  
 (1) 24 (2) 35  
 (3) 40 (4) 48
37. Consider  $N = 32 \times 62 \times 54 = A \times B$ . In how many ways 'N' can be written as a product of two numbers?  
 (1) 16 (2) 24  
 (3) 32 (4) 48
38. Let  $M_x$  be the increasing sequence of natural numbers, each having exactly 24 divisors. Find the single-digit digital root of the smallest such number  $M_1$   
 (1) 1 (2) 3  
 (3) 9 (4) 6
39. Consider  $N = 32 \times 62 \times 54 = A \times B$ . In how many ways 'N' can be written as a product of two different numbers?  
 (1) 16 (2) 24  
 (3) 32 (4) 48
40. A "Pure Factor" of a natural number "n" is that factor of n which has only one prime factor.  $S_n$  is the sum of all the pure factors of n. What is the value of  $S_{3000}$ ?  
 (1) 179 (2) 120  
 (3) 157 (4) 172
41. Consider  $N = 32 \times 62 \times 54 = A \times B$ . In how many ways 'N' can be written as a product of two numbers which are odd?  
 (1) 2 (2) 4  
 (3) 6 (4) 8

42. Let  $N$  be a natural number such that  $N = a \times b$ , where  $a$  and  $b$  are distinct factors of  $N$ . How many such sets of  $(a, b)$  if  $N = 24 \times 33$ ?
- (1) 6 (2) 8  
(3) 10 (4) 12
43. Consider  $N = 32 \times 62 \times 54 = A \times B$ . In how many ways 'N' can be written as a product of two numbers which are even?
- (1) 2 (2) 4  
(3) 6 (4) 8
44. Find the sum of all the natural numbers from 1 to 100 that are neither a multiple of 2 nor a multiple of 5.
- (1) 2000 (2) 2275  
(3) 2515 (4) 2845
45. Consider  $N = 32 \times 62 \times 54 = A \times B$ . In how many ways 'N' can be written as a product of two numbers such that one of them is even and the other is odd?
- (1) 2 (2) 4  
(3) 6 (4) 8
46. Which of the following numbers has exactly eight factors?
- (1) 999 (2) 1001  
(3) 1010 (4) All of these
47. Consider  $N = 32 \times 62 \times 54 = A \times B$ . In how many ways 'N' can be written as a product of two numbers which are co-prime to each other?
- (1) 2 (2) 4  
(3) 6 (4) 8
48. How many factors of 1296 have exactly three factors?
- (1) 1 (2) 2  
(3) 3 (4) 4
49. Find all the prime factors of 660.
- (1) 3, 5, 7 and 13 (2) 2, 3, 5 and 13  
(3) 2, 3, 5 and 7 (4) 2, 3, 5 and 11
50. How many factors of 464 are divisible by 4?
- (1) 2 (2) 4  
(3) 6 (4) 8